

# Saumik Dana

<https://saumikdana.github.io/>

## Summary

US green card holder who builds and operates live ML- and LLM- driven systems end-to-end – research, deployment and infrastructure as one workflow. PhD from The University of Texas at Austin.

## Experience

**Quantitative Researcher, Zebra Capital Management LLC, Stamford CT, Feb 2024 - Dec 2025**

### ML — backed systems

- Built intraday options trading strategies driven by Bayesian anomaly-detection signals
  - Implemented nonlinear optimization and FFT options pricing based stochastic volatility model calibrations
  - Generated *weekly returns of 5 – 10% on peak capital at risk*
- Deployed live equity portfolio with NSGA-II genetic-algorithm optimization and regime-adaptive rebalancing
  - Calibrated Hidden Markov Model backed regime detection and Jensen – Shannon distance based rebalancing
  - Set up optimizer workflow: per-ticker  $\alpha/\beta$  predictions  $\rightarrow$  Bi-objective function  $\rightarrow$  Pareto-optimal weights
  - Achieved *5+% annualized alpha-to-benchmark on multi-year backtest*
- Deployed multi-asset interday options trading strategies using pre-trained models
  - Trained TabPFN classifiers with backtested PnL as target and LoRA-adapted Amazon Chronos based rolling embeddings of technical indicators as features

### LLM — backed Agents

- Built MCP-wrapped tool-calling backed trade signal mining pipeline – Discovered *1.5+ Sharpe strategies*
- Built vision backed (*qwen3p7-plus*) PDF QA system with contrastive-embedding fine-tuning and Qdrant indexing

### LLM — backed systems

- Implemented intraday options trading driven by reasoning (*openai/gpt-oss-120b*) over stochastic volatility signals
  - Designed state-dependent prompt architecture with verdict history and entry-thesis read-back
  - Engineered feature pipeline distilling versioned prompts evaluated over fixed windows — prompt is the strategy
- Deployed news and price action narrative fed intraday FX trading system
  - Built staged pipeline with explicit CoT scaffolding sized to a small non-reasoning model (*llama-3.1-8b-instant*)
  - Closed the loop with DSPy (GEPA) prompt optimization: archived live-runs with price action as rewards
- Deployed news and sovereign yield-curve differentials narrative fed intraday commodity ticker trading system
  - Engineered reasoning (*openai/gpt-oss-120b*) pipeline deriving daily signal from evidence — attribution, conflict-resolution, abstention rules

### Infrastructure

- Designed serverless stacks on AWS (Lambda, EventBridge, DynamoDB, S3, CloudFormation) and Modal
- Engineered GitHub Actions CI/CD pushing digest-pinned ECR images shared across multiple bots
- Handed GitHub secrets to bot-specific YAML templates with DST-aware EventBridge scheduling
- Tuned Lambda for ML inference with multi-GB ephemeral storage and S3-backed HuggingFace/Torch caches
- Built React/static dashboards over DynamoDB/S3, served via FastAPI behind Lambda Function URLs

**Computational Engineer, VISIE Inc., Austin TX, Aug 2023 - Nov 2023**

- Joined during early integration of imaging and robotic actuation in surgical navigation platform
- Implemented TCP/UDP communication protocols to control robotic arm motion
- Supported product packaging, deployment, live demonstration used in *successful Series A round*

**Computational Lead, Sophelio, Austin TX, Aug 2022 - Mar 2023**

- Adapted physics-informed modeling—originally for fusion experiment data—to financial time series
- Engineered production pipeline: sparse regression for PDE construction, signal generation
- Implemented CAGR maximizing Bayesian TPE optimizer based swing trading system

## Education

**PhD, Engineering Mechanics — University of Texas at Austin, Austin TX**

**MEng, Mechanical Engineering — Indian Institute of Science, Bangalore, India**

**BEng, Mechanical Engineering — University of Mumbai, Mumbai, India**

## Technical Skills

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Languages/Data — Python, pandas, Polars, NumPy, requests, BeautifulSoup, RSS/XML

Cloud — AWS (Lambda, DynamoDB, S3, ECR, EventBridge, API Gateway, CloudFormation), Modal, Docker

LLM — Groq, Langchain, Fireworks AI, Qdrant, DSPy, MCP, Tool calling

APIs — FastAPI, React/JSX, Next.js, HTML, Alpaca, OANDA, TradeStation, yfinance

ML/Quant — PyTorch, scikit-learn, TabPFN, Chronos, PEFT, hmmlearn, SciPy, pymoo, autograd